Principal Component Analysis (PCA)

Complete Step-by-Step Solved Numerical Example

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Theoretical Optimization Objective

- **The Goal:** Given a data matrix X, find a directional unit vector v ($v^T v = 1$) that maximizes the variance of the projected data points (Xv).
- **Variance Matrix Formulation:**

$$\text{Var}(Xv) = \frac{1}{N-1}(Xv)^T(Xv) = v^T \left(\frac{X^T X}{N-1} \right) v = v^T C v$$

- **Constrained Optimization:** Maximize variance under a unit constraint:

$$\max_{v} \quad v^T C v \quad \text{subject to} \quad v^T v = 1$$

- **The Lagrangian Proof:** Constructing $\mathcal{L}(v, \lambda) = v^T C v - \lambda(v^T v - 1)$ and setting $\frac{\partial \mathcal{L}}{\partial v} = 0$ yields:

$$Cv = \lambda v$$

- **Core Takeaway:** The projection axis v **must be an eigenvector** of the covariance matrix C, and the variance along that axis equals its eigenvalue (λ).

Step-by-Step Solution Algorithm

Step 1: Mean Centering

Subtract the calculated mean ($\bar{X}_i$) of each feature from every data point to center the dataset around the coordinate origin ($\mu = 0$).

Step 2: Compute Covariance Matrix (C)

Calculate the joint variability between features using sample degrees of freedom ($N - 1$):

$$\text{Cov}(X_i, X_j) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{ik} - \bar{X}_i)(x_{jk} - \bar{X}_j)$$

Step 3: Solve Characteristic Equation

Isolate the scalar roots (λ) and their matching raw directional eigenvectors (u):

$$\det(C - \lambda I) = 0 \quad \text{and} \quad (C - \lambda I)u = 0$$

Step-by-Step Solution Algorithm (Cont.)

Step 4: Normalize Eigenvectors

Standardize the length of each structural eigenvector to a unit length of 1:

$$\text{Length} = \sqrt{u_1^2 + \cdots + u_n^2} \implies E_i = \frac{u}{\text{Length}}$$

Step 5: Component Selection & Projection

- Sort eigenvalues and normalized eigenvectors in descending order ($\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$).
- Keep the top p eigenvectors to form your transformation matrix.
- Matrix-multiply this transformation matrix by your centered data matrix to compress the coordinates into lower dimensions.

Problem Statement

Table: Dataset Array

Sample Index	1	2	3	4	5	6	7	8	9	10
Original X	2.5	0.5	2.2	1.9	3.1	2.3	2.0	1.0	1.5	1.1
Original Y	2.4	0.7	2.9	2.2	3.0	2.7	1.6	1.1	1.6	0.9

- **Dataset:** Given a 2D dataset with two features (X and Y) and 10 data points ($N = 10$).
- **Objective:** Apply the PCA algorithm to reduce the dimensionality from 2D to 1D ($n = 2 \rightarrow p = 1$).
- **Step 0: Find Feature Means**
  - $\bar{X} = \frac{\sum X}{N} = \frac{2.5+0.5+2.2+1.9+3.1+2.3+2.0+1.0+1.5+1.1}{10} = \frac{18.1}{10} = \mathbf{1.81}$
  - $\bar{Y} = \frac{\sum Y}{N} = \frac{2.4+0.7+2.9+2.2+3.0+2.7+1.6+1.1+1.6+0.9}{10} = \frac{19.1}{10} = \mathbf{1.91}$

Step 1: Constructing the Deviation and Product Tables

No.	X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
1	2.5	2.4	0.69	0.49	0.4761	0.2401	0.3381
2	0.5	0.7	-1.31	-1.21	1.7161	1.4641	1.5851
3	2.2	2.9	0.39	0.99	0.1521	0.9801	0.3861
4	1.9	2.2	0.09	0.29	0.0081	0.0841	0.0261
5	3.1	3.0	1.29	1.09	1.6641	1.1881	1.4061
6	2.3	2.7	0.49	0.79	0.2401	0.6241	0.3871
7	2.0	1.6	0.19	-0.31	0.0361	0.0961	-0.0581
8	1.0	1.1	-0.81	-0.81	0.6561	0.6561	0.6561
9	1.5	1.6	-0.31	-0.31	0.0961	0.0961	0.0961
10	1.1	0.9	-0.71	-1.01	0.5041	1.0201	0.7171
Σ	**18.1**	**19.1**	**0.00**	**0.00**	**5.5490**	**6.4490**	**5.5390**

Step 1 Continued: Finding Covariance Matrix Elements

- Formula for Sample Covariance uses divisor $(N-1) = 10-1 = 9$:

$$\text{Cov}(X,X) = \frac{\sum(X-\bar{X})^2}{N-1} = \frac{5.5490}{9} = \mathbf{0.6166}$$

$$\text{Cov}(Y,Y) = \frac{\sum(Y-\bar{Y})^2}{N-1} = \frac{6.4490}{9} = \mathbf{0.7166}$$

$$\text{Cov}(X,Y) = \text{Cov}(Y,X) = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{N-1} = \frac{5.5390}{9} = \mathbf{0.6154}$$

- Combined into Covariance Matrix C:

$$C = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}$$

Step 2: Solving for Eigenvalues (λ)

- Characteristic Matrix Formula: $\det(C - \lambda I) = 0$

$$\det \begin{bmatrix} 0.6166 - \lambda & 0.6154 \\ 0.6154 & 0.7166 - \lambda \end{bmatrix} = 0$$

- Expand the cross-multiplication determinant equation:

$$(0.6166 - \lambda)(0.7166 - \lambda) - (0.6154 \times 0.6154) = 0$$

$$(\lambda^2 - 0.7166\lambda - 0.6166\lambda + 0.4419) - 0.3787 = 0$$

$$\lambda^2 - 1.3332\lambda + 0.0632 = 0$$

- Solving the quadratic system isolates our roots:
  - $\lambda_1 = 0.049$ (Minor variation path)
  - $\lambda_2 = 1.284$ (Principal variance direction)

Step 3: Calculating Eigenvector V_1 for $\lambda_1 = 0.049$

- Substitute $\lambda_1 = 0.049$ into matrix format $(C - \lambda_1 I)V_1 = 0$:

$$\begin{bmatrix} 0.6166 - 0.049 & 0.6154 \\ 0.6154 & 0.7166 - 0.049 \end{bmatrix} \begin{bmatrix} X_1 \\ Y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies$$

$$\begin{bmatrix} 0.5676 & 0.6154 \\ 0.6154 & 0.6676 \end{bmatrix} \begin{bmatrix} X_1 \\ Y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- Multiplying out the first row yields: $0.5676X_1 + 0.6154Y_1 = 0$
- Assume $Y_1 = 1 \implies 0.5676X_1 = -0.6154 \implies X_1 = \mathbf{-1.084}$
- Raw Eigenvector $V_1 = [-1.084, 1]^T$. Find vector length to normalize:

$$\text{Length} = \sqrt{(-1.084)^2 + (1)^2} = \sqrt{1.175 + 1} = \sqrt{2.175} \approx \mathbf{1.475}$$

- **Normalized Vector V_1:**

$$V_1 = \begin{bmatrix} -1.084/1.475 \\ 1/1.475 \end{bmatrix} = \begin{bmatrix} -0.735 \\ 0.678 \end{bmatrix}$$

Step 3 Continued: Calculating Eigenvector V_2 for $\lambda_2 = 1.284$

- Substitute $\lambda_2 = 1.284$ into matrix format $(C - \lambda_2 I)V_2 = 0$:

$$\begin{bmatrix} 0.6166 - 1.284 & 0.6154 \\ 0.6154 & 0.7166 - 1.284 \end{bmatrix} \begin{bmatrix} X_2 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies$$

$$\begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix} \begin{bmatrix} X_2 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- Multiplying out the first row yields: $-0.6674X_2 + 0.6154Y_2 = 0$
- Assume $Y_2 = 1 \implies -0.6674X_2 = -0.6154 \implies X_2 = \mathbf{0.922}$
- Raw Eigenvector $V_2 = [0.922, 1]^T$. Find vector length to normalize:

$$\text{Length} = \sqrt{(0.922)^2 + (1)^2} = \sqrt{0.850 + 1} = \sqrt{1.850} \approx \mathbf{1.360}$$

- **Normalized Vector V_2 (Principal Component Axis):**

$$V_2 = \begin{bmatrix} 0.922/1.360 \\ 1/1.360 \end{bmatrix} = \begin{bmatrix} 0.677 \\ 0.735 \end{bmatrix}$$

Step 4: Choosing Component Axis & Building Feature Vector

- Since λ_2 (1.284) $> \lambda_1$ (0.049), λ_2 forms our dominant projection component axis.
- **Dimensional Projection Formula:**

$$\text{New Feature Point } (z_i) = (\text{Centered } X_i \times 0.677) + (\text{Centered } Y_i \times 0.735)$$

- Let's calculate the transformations row-by-row:
  - $z_1 = (0.69 \times 0.677) + (0.49 \times 0.735) = 0.467 + 0.360 = \mathbf{0.827}$
  - $z_2 = (-1.31 \times 0.677) + (-1.21 \times 0.735) = -0.887 - 0.890 = \mathbf{-1.777}$
  - $z_3 = (0.39 \times 0.677) + (0.99 \times 0.735) = 0.264 + 0.728 = \mathbf{0.992}$
  - $z_4 = (0.09 \times 0.677) + (0.29 \times 0.735) = 0.061 + 0.213 = \mathbf{0.274}$
  - $z_5 = (1.29 \times 0.677) + (1.09 \times 0.735) = 0.873 + 0.801 = \mathbf{1.674}$

Step 4 Continued: Final 1D Transformed Feature Vector

- Computing rows 6 to 10 using the structural vector multiplier properties:
  - $z_6 = (0.49 \times 0.677) + (0.79 \times 0.735) = 0.332 + 0.581 = \mathbf{0.913}$
  - $z_7 = (0.19 \times 0.677) + (-0.31 \times 0.735) = 0.129 - 0.228 = \mathbf{-0.099}$
  - $z_8 = (-0.81 \times 0.677) + (-0.81 \times 0.735) = -0.548 - 0.595 = \mathbf{-1.143}$
  - $z_9 = (-0.31 \times 0.677) + (-0.31 \times 0.735) = -0.210 - 0.228 = \mathbf{-0.438}$
  - $z_{10} = (-0.71 \times 0.677) + (-1.01 \times 0.735) = -0.481 - 0.742 = \mathbf{-1.223}$
- **Conclusion Summary:** The original 2-column spatial arrangement configuration matrix has been compressed entirely into a singular 1-dimensional array list column.